

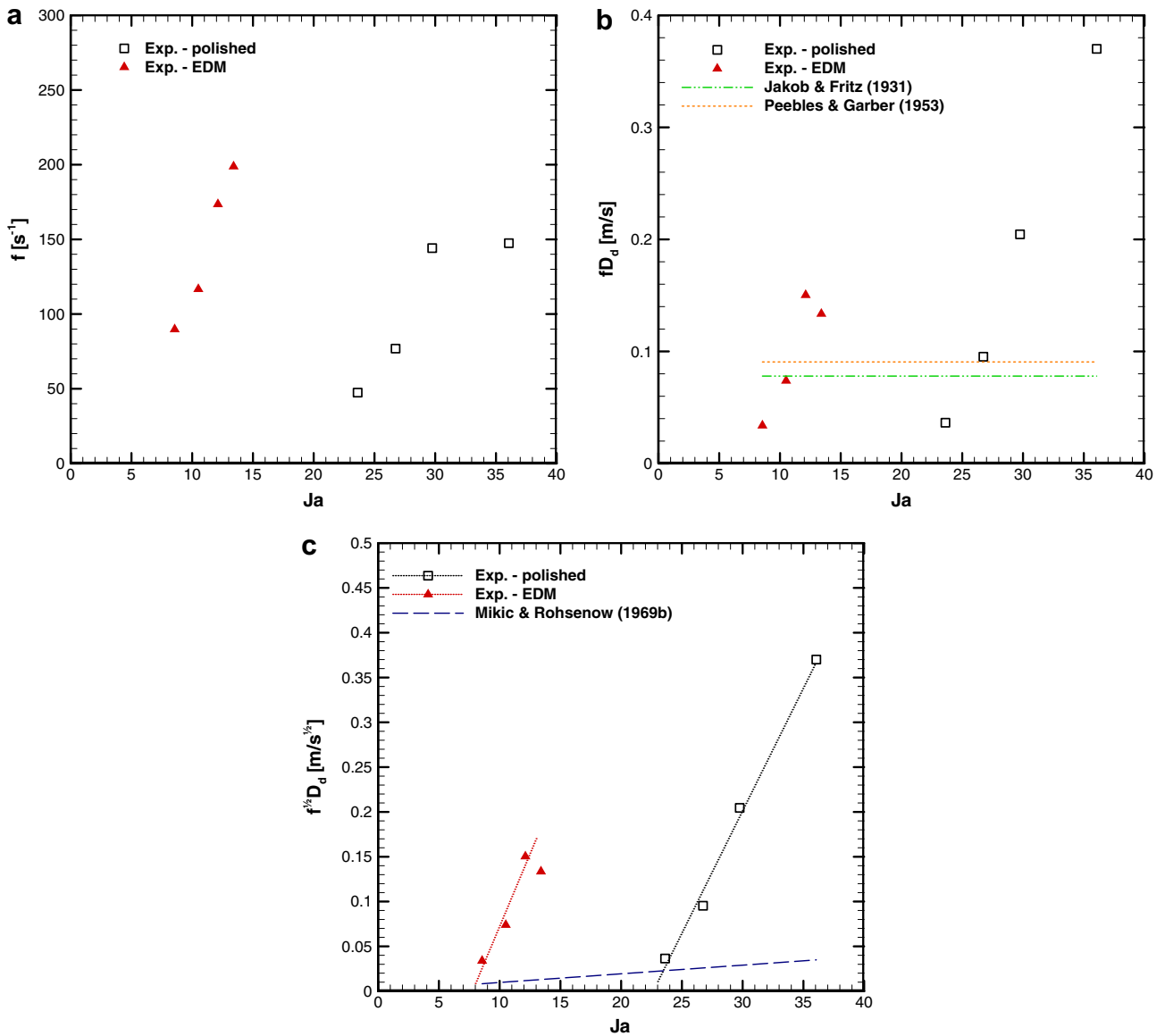


Fig. 13. Measured bubble departure (a) frequency, (b) frequency–diameter product, and (c) root-frequency–diameter product, compared with some commonly used correlations.

$$fD_d = 1.18 \left(\frac{t_g}{t_g + t_w} \right) \left(\frac{\sigma g (\rho_l - \rho_v)}{\rho_l^2} \right)^{0.25} \quad (3)$$

of growth period to bubbling period, while Eq. (4) also accounts for wall superheat relation in terms of Ja . Eq. (2) was developed based on experimental data from the pool boiling of water and li